

Worked Answer Key

Page 1

1. A line extends forever in both directions; a segment has two endpoints; a ray has one endpoint and extends forever in one direction. Examples: line $\overleftrightarrow{AB}$, segment $\overline{CD}$, ray $\overrightarrow{EF}$.
2. The segment has endpoints A, B ; the ray starts at C and passes through D ; the line has arrows in both directions through E, F .

Page 2

1. Parallel lines never meet and stay the same distance apart. Perpendicular lines meet at a 90° angle.
2. They are parallel because opposite sides of a rectangle run in the same direction; they are not perpendicular to each other.

Page 3

1. 35° acute; 90° right; 125° obtuse; 180° straight.
2. Answers vary. Draw and label one acute angle (between 0° and 90°), one right angle (90°), one obtuse angle (between 90° and 180°), and one straight angle (180°). The measures from problem 1 are valid examples, but other acute and obtuse measures are also correct.

Page 4

1. A correct protractor construction has two rays sharing a vertex and measure 65° ; it is acute. For example, label the vertex V , a point A on one ray, and a point B on the other; the rays are $\overrightarrow{VA}$ and $\overrightarrow{VB}$.
2. A correct construction measures 140° ; it is obtuse because it is between 90° and 180° .

Page 5

1. $180 - 72 = 108^\circ$.
2. $90 - 37 = 53^\circ$.

Page 6

1. Equilateral: all three sides are equal.
2. Isosceles: exactly two sides are equal (5 in and 5 in).

Page 7

1. Right triangle: one angle is 90° .
2. Acute triangle: all three angles are less than 90° .

Page 8

1. Square; it has four equal sides and four right angles.
2. Trapezoid; draw one pair of parallel sides and one pair that is not parallel.

Page 9

1. Heptagon; it has 7 vertices.
2. Triangle—3; pentagon—5; hexagon—6; octagon—8.

Page 10

1. $P = 2(12 + 7) = 38$ cm.
2. $P = 5 \times 9 = 45$ m.

Page 11

1. $A = 14 \times 6 = 84$ square feet.
2. $\text{Area} = 11 \times 11 = 121$ square inches; $\text{perimeter} = 4 \times 11 = 44$ inches.

Page 12

1. $8 \times 3 + 4 \times 5 = 24 + 20 = 44$ square meters.
2. $10 \times 9 - 3 \times 4 = 90 - 12 = 78$ square units.

Page 13

1. From the origin, plot A 2 units right and 4 up, B 6 right and 4 up, and C 6 right and 1 up. A and B share a horizontal line because both have $y = 4$.
2. $(5, 3)$: the first coordinate is 5 right; the second is 3 up.

Page 14

1. Both points have $y = 2$; $7 - 1 = 6$ units.
2. The shape is a rectangle. Side lengths are 4 and 5, so $P = 2(4 + 5) = 18$ units.

Page 15

1. A square has 4 lines of symmetry: two through opposite vertices and two through opposite side midpoints.
2. A non-square rectangle has 2 lines of symmetry, through the midpoints of opposite sides. Fold it in half so the left and right edges meet, or so the top and bottom edges meet; each fold makes the halves match exactly. A diagonal fold does not make the halves match unless the rectangle is a square.

Page 16

1. An equilateral triangle has 3 lines of symmetry, one from each vertex to the midpoint of the opposite side.
2. A regular pentagon has 5 lines of symmetry, one through each vertex and the midpoint of the opposite side. Rule: a regular polygon has as many lines of symmetry as sides.

Page 17

1. $(5, 2)$. The fold line is 4 units right of the origin. The mirror point is $4 + 1 = 5$ units right of the origin and remains 2 units up. Both points are 1 unit from the fold line on opposite sides.
2. $(4, 5)$. The fold line is 3 units up. The mirror point is $3 + 2 = 5$ units up and remains 4 units right. Both points are 2 units from the fold line on opposite sides.

Page 18

1. No. It is a rhombus because it is a quadrilateral with four equal sides. A square must also have four right angles.
2. It is a rectangle. It could be a square only if all four sides were also equal.

Page 19

1. Perimeter $= 2(18 + 9) = 54$ m. Area $= 18 \times 9 = 162$ square meters.
2. A regular hexagon has 6 equal sides: $48 \div 6 = 8$ cm.

Page 20

1. It is a right angle. Two perpendicular lines intersect to form four right angles.
2. Area $= 8 \times 5 = 40$ square units; perimeter $= 2(8 + 5) = 26$ units; a non-square rectangle has 2 lines of symmetry.